

Inverse Probability Weighting

Weak Paradoxes, Evidence Estimation, and PPI

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Outline

Inverse Probability Weighting: Weak Paradoxes, Evidence Estimation, and PPI[1]

Outline of this talk

1. **IPW estimators & the weak paradoxes:** HT vs. Hájek, Robins-Ritov-Wasserman, and the Bayesian answers as *Stein shrinkage*.
2. **Prediction-powered inference (PPI):** The PPI rectifier *is* a Hájek estimator; IPW-adjusted PPI under informative labeling.
3. **Evidence estimation:** Importance sampling as IPW; quantile importance sampling and the $O(n^{-4})$ rate.

[1] Based on a series of papers (NEJSDS 2026; Stat. Sci. Short Comm. 2026; BJPS 2025).

Inverse Probability Weighting and the Weak Paradoxes

Horvitz–Thompson and Hájek

Finite population $U = \{1, \ldots, N\}$ with values Y_k; sample $s \subset U$ with inclusion probabilities $p_k = \mathbb{P}(k \in s)$. Target: $\psi = N^{-1} \sum_{k \in U} Y_k$.

$$\hat{\psi}_{\mathrm{HT}} = \frac{1}{N} \sum_{k \in s} \frac{Y_k}{p_k} \qquad \text{(unbiased; Horvitz and Thompson 1952)}$$

$$\hat{\psi}_{\mathrm{Há}} = \frac{\sum_{k \in s} Y_k / p_k}{\sum_{k \in s} 1 / p_k} \qquad \text{(normalized; needs no } N\text{; Hájek 1971)}$$

- HT is unbiased, admissible and consistent [Ramakrishnan, 1973, Delevoye and Sävje, 2020].
- These are **design-based** weights: unit k "represents" $1/p_k$ units of the population [Neyman, 1934, Little, 2008].

Missing-data view and adaptive normalization

An equivalent framework arises in the context of missing data, *i.e.*, when subjects are observed with nonuniform probabilities.

Bernoulli response indicators $R_k \overset{ind}{\sim} \text{Bernoulli}(p_k)$, i.e., R_k indicates whether Y_k were observed or not.

We could write:

$$\hat{S} = \sum_{k=1}^{n} \frac{Y_k R_k}{p_k}, \quad \hat{n} = \sum_{k=1}^{n} \frac{R_k}{p_k} \tag{1}$$

$$\hat{\psi}_{HT} = \frac{\hat{S}}{n}, \quad \hat{\psi}_{\text{Hájek}} = \frac{\hat{S}}{\hat{n}}. \tag{2}$$

Trotter and Tukey [1956]

CONDITIONAL MONTE CARLO FOR NORMAL SAMPLES+

Hale F. Trotter and John W. Tukey

Princeton University

The techniques presented here represent what sometimes happens to lazy people who start doing a computation before they quite know what they are going to do. Contrary to experience and high moral principles, this time it worked out all right. We solved our problem (cp. "Monte Carlo techniques in a complex problem about normal samples" pp. 80 ff., below) and we were able to extract some relatively general techniques from what we were driven to do in attempting to solve a particular problem.

7. If we have N weighted samples (y_1, w_1), (y_2, w_2), ..., (y_N, w_N) from some distribution and are interested in ave $\left[\phi(z) \mid \text{distribution}\right]$ we can estimate the average either by

$$\frac{\Sigma w_i \phi(y_i)}{N} \quad \text{or by} \quad \frac{\Sigma w_i \phi(y_i)}{\Sigma w_i}$$

Experience shows that the former is almost always better than the latter (as well as being unbiased).

8. The equivalence for large samples of these two estimates follows from 5(b) by taking $\phi(z) \equiv 1$, whence

$$\text{ave}\left[w \mid \text{process}\right] = 1.$$

Initially we didn't realize how important it was to distinguish these two estimates -- but experience showed the advantage of dividing by N, the average total weight, rather than by the realized total weight, Σw_i. Indeed, it is possible to use

$$\frac{\Sigma w_i \phi(y_i)}{\lambda N + (1 - \lambda)\Sigma w_i}$$

for any real λ, and there is some ground for anticipating that λ's somewhat larger than unity will often be best. (We don't know of any practical experience with such more general estimates.)

Adaptive Normalization

- **Trotter–Tukey** [Trotter and Tukey, 1956] interpolates between them:

$$\hat{\psi}_{TT} = \frac{\hat{S}}{(1-\lambda)n + \lambda\hat{n}}, \qquad \lambda \in \mathbb{R}.$$

- **Adaptive normalization** [Khan and Ugander, 2023] lets the data pick $\hat{\lambda}$:

$$\mathbb{V}(\hat{\psi}_{AN}) \;\leq\; \min\{\mathbb{V}(\hat{\psi}_{\mathrm{HT}}),\, \mathbb{V}(\hat{\psi}_{\mathrm{Há}})\}.$$

- A recurring theme: *self-normalization / adaptivity* achieves stability at the price of a small bias.

Weak Paradoxes

Weak paradoxes

"Many years ago I argued, tongue in cheek, that since Dr. D. Basu has found no paradox residing in the Bayesian paradigm, may be the paradigm has no paradox. I do not mean a paradox in the strict mathematical sense of leading to a logical inconsistency in the paradigm but rather an example which doesn't, strictly speaking, violate logic but nonetheless makes us uneasy and embarrassed. I call an example like this a weak paradox." - [Ghosh, 2015]

Weak Paradoxes and Paradigms

Jayanta K. Ghosh

Purdue University, USA and Indian Statistical Institute, India

We consider two important counterexamples, one due to Basu (1988) and the other due to Wasserman (2004). We argue that while they provide insights about different weaknesses in Frequentist and Bayesian paradigms, too much stress on unbiasedness in the former and the full likelihood in the latter, the counterexamples are not as negative as appeared at first sight. It is shown how the paradigms can deal with the counterexamples without any fundamental change.

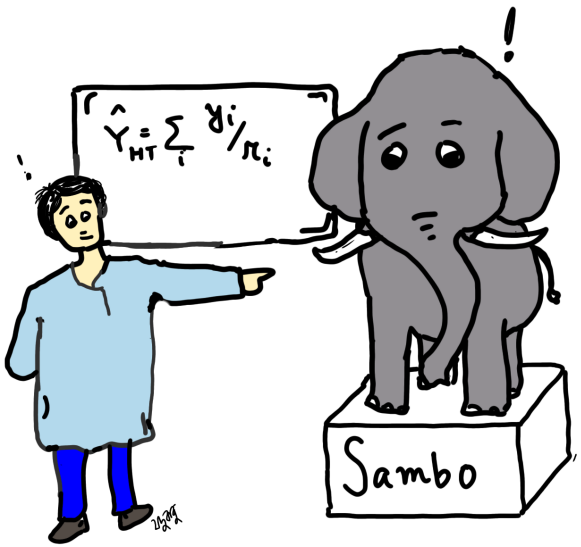
Keywords: Bayesian and Frequentist paradigms, Counterexamples, Pivotal variables, Unequal probability sampling.

D. Basu and the logical foundations of sampling

- Basu's *"An essay on the logical foundations of survey sampling"* placed the **likelihood principle** at the centre of the design-vs-model debate (Godambe–Basu).
- His verdict on **Horvitz–Thompson**-style weighting: design-based weights $w_k \propto 1/p_k$ can produce estimators that are unbiased and admissible, yet *absurd* on a given sample.
- Hájek [1971], commenting on Basu's essay, proposed a **ratio-normalised** estimator, the Hájek estimator, as the more sensible choice.

Basu's circus

Example 3. The circus owner is planning to ship his 50 adult elephants and so he needs a rough estimate of the total weight of the elephants. As weighing an elephant is a cumbersome process, the owner wants to estimate the total weight by weighing just one elephant. Which elephant should he weigh? So the owner looks back on his records and discovers a list of the elephants' weights taken 3 years ago. He finds that 3 years ago Sambo the middle-sized elephant was the average (in weight) elephant in his herd. He checks with the elephant trainer who reassures him (the owner) that Sambo may still be considered to be the average elephant in the herd. Therefore, the owner plans to weigh Sambo and take $50y$ (where y is the present weight of Sambo) as an estimate of the total weight $Y = Y_1 + \ldots + Y_{50}$ of the 50 elephants. But the circus statistician is horrified when he learns of the owner's purposive samplings plan. "How can you get an unbiased estimate of Y this way?" protests the statistician. So, together they work out a compromise sampling plan. With the help of a table of random numbers they devise a plan that allots a selection probability of 99/100 to Sambo and equal selection probabilities of 1/4900 to each of the other 49 elephants. Naturally, Sambo is selected and the owner is happy. "How are you going to estimate Y?", asks the statistician. "Why? The estimate ought to be $50y$ of course," says the owner. "Oh! No! That cannot possibly be right," says the statistician, "I recently read an article in the *Annals of Mathematical Statistics* where it is proved that the Horvitz-Thompson estimator is the unique hyperadmissible estimator in the class of all generalized polynomial unbiased estimators." "What is the Horvitz-Thompson estimate in this case?" asks the owner, duly impressed. "Since the selection probability for Sambo in our plan was 99/100," says the statistician, "the proper estimate of Y is $100y/99$ and not $50y$." "And, how would you have estimated Y," inquires the incredulous owner, "if our sampling plan made us select, say, the big elephant Jumbo?" "According to what I understand of the Horvitz-Thompson estimation method," says the unhappy statistician, "the proper estimate of Y would then have been $4900y$, where y is Jumbo's weight." That is how the statistician lost his circus job (and perhaps became a teacher of statistics!)



Basu's elephants

- I slightly prefer C. A. Sims' retelling, and I'll present his version with minor tweaks [2]
- A circus owner needs to estimate the total weight of his ten elephants, to decide how many train cars he will need to transport them.
- He discusses the problem with his statistician: "I'm thinking of just weighing Sambo and multiplying by 10."
- The statistician points out that this would be a biased estimate, and suggests an unbiased one: pick an elephant at random, weigh that one, and multiply by 10.
- The circus owner points out that he proposed weighing Sambo because he is easy to handle and trained to the scale; the other elephants would be much riskier to try to weigh.

[2] *Large Parameter Spaces and Weighted Data: A Bayesian Perspective*, Laplace's Demon seminar (BMLS), 28 Feb 2022. [video]

The statistician's solution

- The statistician sees that this calls for lowering the cost, while preserving unbiasedness:
- "Choose Sambo with probability .99, and if not him choose one of the remaining 9 with equal probabilities."
- The circus owner likes this. "I almost surely do exactly what I originally proposed, weigh Sambo and multiply by 10, with just a slight risk of incurring heavier cost, and yet now I have a statistically sound, unbiased estimate!"

Statistician explains the owner's error

- "You're right that with high probability you weigh Sambo, but you multiply his weight by 1.01, not 10."
- Owner: "But there are 10 elephants!" *(pause for thought)* "What if I end up having to weigh a different elephant, say Jumbo?"
- Statistician: "In that case, you multiply the weight by 900."
- Circus owner refuses to pay the consulting fee.
- In the original version, Basu wrote, "that is how the Statistician lost his circus job (and perhaps became a teacher of Statistics!)."

When is Hájek preferable?

Särndal et al. [2003] identify three regimes where $\mathbb{V}(\hat{\psi}_{\text{Há}}) < \mathbb{V}(\hat{\psi}_{\text{HT}})$:

1. the y_k are close to ψ (e.g. nearly constant);
2. variable sample size with (near-)equal inclusion probabilities;
3. p_k **negatively correlated** with y_k, i.e., large y_k, small p_k.

Condition (iii) is Basu's circus example

When y_k / p_k fluctuates wildly, HT is unstable; but Hájek adapts. An auxiliary $A_k \propto Y_k$ with known total leads to the Hájek's ratio estimator:

$$\hat{Y}_{\text{Há}} = \left(\sum_k A_k \right) \frac{\sum_k Y_k / p_k}{\sum_k A_k / p_k},$$

immune to the circus example pathology. Setting $A_k \equiv 1$ recovers the Hájek mean.

Robins–Ritov–Wasserman example: setup

A high-dimensional missing-data problem [Robins and Ritov, 1997, Wasserman, 2004].

IID triples (X_i, R_i, Y_i), $i = 1, \ldots, n$, with $B \gg n$ (think $B = 100^{100}$):

$$X_i \sim \mathcal{U}\{1, \ldots, B\}, \quad R_i \mid X_i \sim \mathrm{Bern}(p_{X_i}), \quad Y_i \mid R_i{=}1, X_i \sim \mathrm{Bern}(\theta_{X_i}).$$

- Cell probabilities $\theta = (\theta_1, \ldots, \theta_B)$ **unknown**; propensities $p = (p_1, \ldots, p_B)$ **known**, bounded $0 < \delta \le p_j \le 1 - \delta < 1$.
- $R_i = 0$ means "Y_i missing." Target: the population prevalence

$$\psi = \mathbb{P}(Y_i = 1) = \tfrac{1}{B} \textstyle\sum_{b=1}^{B} \theta_b = g(\theta).$$

Issue

ψ is a functional of a B-dimensional θ, but each draw informs *at most one* coordinate. The design that governs missingness lives *outside* the likelihood.

Robins–Ritov–Wasserman example

The likelihood is easy to write down, but there is very little information on most θ_j's.

So Bayes estimates of the θ_j's would be poor, and taking their average gives a Bayes estimate for ψ that is unlikely to be consistent.

On the other hand, a **Horvitz–Thompson** estimate, namely

$$\hat{\psi} = \frac{1}{n} \sum_{i=1}^{n} \frac{R_i Y_i}{p_{X_i}},$$

is unbiased and has variance $\leq \frac{1}{n\delta^2}$.

This implies $\hat{\psi}$ is consistent because it is unbiased and the conditions on p_{X_i}'s. But "it cannot be derived from a Bayesian or likelihood point of view" ... since the p_{X_i}'s "drop out of the log-likelihood" [Wasserman, 2004].

Why the likelihood "goes awry"

The likelihood for a single observation factorizes, and the design parameters are *known constants*:

$$f(X_i, R_i, Y_i) \propto \underbrace{\frac{1}{B}}_{\text{known}} \quad \underbrace{p_{X_i}^{R_i}(1-p_{X_i})^{1-R_i}}_{\text{known} \Rightarrow \text{ancillary for } \theta} \quad \theta_{X_i}^{Y_i R_i}(1-\theta_{X_i})^{(1-Y_i)R_i}.$$

So the whole-sample likelihood keeps *only* the outcome factors:

$$\mathcal{L}(\theta) \propto \prod_{j=1}^{B} \theta_j^{n_j}(1-\theta_j)^{m_j},$$

where $n_j = \#\{i : Y_i{=}1, R_i{=}1, X_i{=}j\}$ and $m_j = \#\{i : Y_i{=}0, R_i{=}1, X_i{=}j\}$ count the observed successes and failures in cell j.

Consequences of $B \gg n$

Since $\leq n$ of the B cells are ever sampled, $n_j = m_j = 0$ for *almost all* j. For those cells θ_j never enters $\mathcal{L}$, so its posterior equals its prior. As ψ averages over all B cells, $f(\psi \mid \text{data}) \approx f(\psi)$: under an **independent** prior the data barely update ψ.

The paradox

Frequentist fix. The Horvitz–Thompson estimator uses the known p_{X_i}:

$$\hat{\psi}_{\mathrm{HT}} = \frac{1}{n}\sum_{i=1}^{n}\frac{R_i Y_i}{p_{X_i}}, \qquad \mathbb{E}(\hat{\psi}_{\mathrm{HT}}) = \psi, \qquad \mathbb{V}(\hat{\psi}_{\mathrm{HT}}) \leq \frac{1}{n\delta^2}.$$

MSE is $O(1/n)$ *for every* B: consistency with no restriction on dimension.

If p_{X_i}'s are not properly aligned with $R_i Y_i$, unbiasedness can force the variance to explode, as in the circus example.

Wasserman

The winning estimator is built from the propensities p_{X_i} - the very terms that are *ancillary* and vanish from the likelihood, so no likelihood- or Bayes-based rule with an independent prior can use them.

"Bayesians are slaves to the likelihood function. . . when the likelihood goes awry, so will Bayesian inference" [Wasserman, 2004, p.189].

Why "weak." The paradox dissolves once the prior stops treating the θ_b as independent: an *exchangeable* prior lets observed cells inform unobserved ones: Stein-type shrinkage.

Bayesian answer I: exchangeability and shrinkage [Li, 2010]

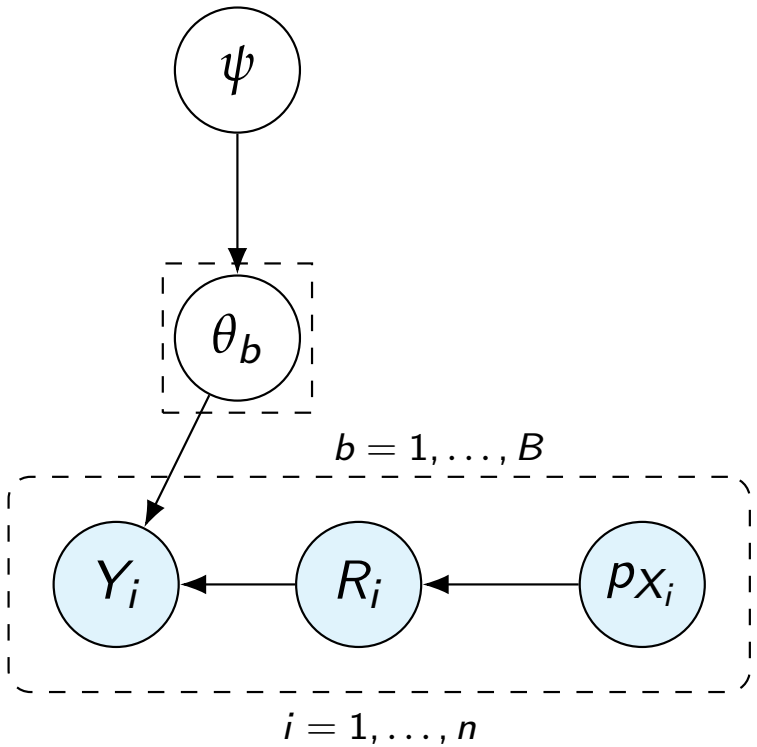
Make the θ_b **exchangeable**, not independent:

$$\theta_b \mid \alpha_T, \psi \overset{ind}{\sim} \text{Beta}\big(\alpha_T \psi,\, \alpha_T(1-\psi)\big)$$
$$(\psi, \alpha_T) \mid \alpha_F \sim \text{Beta}(\alpha_F, \alpha_F) \times \pi(\alpha_T)$$

Posterior mean collapses to a shrinkage estimator:

$$\hat{\psi}_{\text{Li}} = \frac{\sum_i R_i Y_i + \alpha_F}{\sum_i R_i + 2\alpha_F}.$$

Key: the p_b vanish from the denominator $\Rightarrow$ *no* δ restriction needed.



Hierarchical model for the RRW problem.

This is **Stein's phenomenon**: moving from independence to exchangeability *borrows strength* across the θ_b, trading a little bias for a large variance reduction [Ghosh, 2015].

Properties of Li's estimator: when does Bayes stay consistent?

A delta-method (linearization) analysis of $\hat{\psi}_{\mathrm{Li}} = (\sum_i R_i Y_i + \alpha_F)/(\sum_i R_i + 2\alpha_F)$ gives, with $\bar{\theta}_{\cdot p} = \frac{1}{B}\sum_b \theta_b p_b$ and $\sigma_{\theta,p} = \bar{\theta}_{\cdot p} - \bar{p}\,\psi$,

$$\mathbb{E}(\hat{\psi}_{\mathrm{Li}}) \;\longrightarrow\; \psi + \frac{\sigma_{\theta,p}}{\bar{p}}, \qquad \mathbb{V}(\hat{\psi}_{\mathrm{Li}}) = O(1/n).$$

Consistency dichotomy

- **MCAR** ($\theta_X \perp p_X \mid X$, i.e. $\sigma_{\theta,p} = 0$): $\hat{\psi}_{\mathrm{Li}}$ is consistent, $O(1/n)$ variance, *no* δ bound needed.
- **MAR / confounding** ($\sigma_{\theta,p} \neq 0$): $\hat{\psi}_{\mathrm{Li}}$ carries an asymptotic bias $\sigma_{\theta,p}/\bar{p}$: Li's estimator is consistent only if $\mathbb{E}(Y \mid R{=}1) = \mathbb{E}(Y)$ [Ritov et al., 2014, see also].

Linero [2024]: such "innocuous" priors encode an a-priori belief that selection bias is minimal.

Bayesian answer II: binning–smoothing

Weak Paradoxes and Paradigms

Jayanta K. Ghosh

Purdue University, USA and Indian Statistical Institute, India

We consider two important counterexamples, one due to Basu (1988) and the other due to Wasserman (2004). We argue that while they provide insights about different weaknesses in Frequentist and Bayesian paradigms, too much stress on unbiasedness in the former and the full likelihood in the latter, the counterexamples are not as negative as appeared at first sight. It is shown how the paradigms can deal with the counterexamples without any fundamental change.

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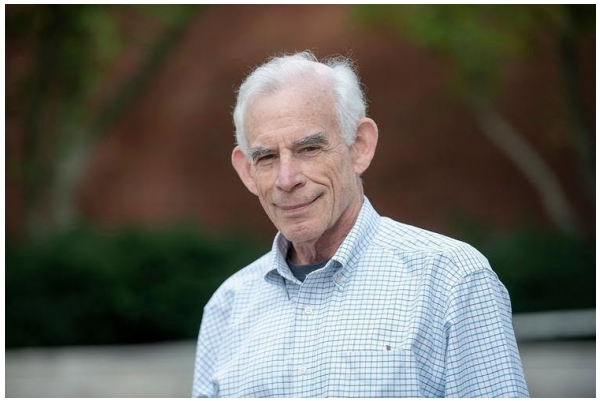
Binning–smoothing [Ghosh, 2015]

Partition $(\delta, 1-\delta)$ into $k \ll B$ bins; replace each p_{X_i} by the bin mean $\tilde{p}_j$:

$$\hat{\psi}_{\mathrm{BS:HT}} = \sum_{j=1}^{k} \frac{n_j}{n} \frac{1}{n_j} \frac{\sum_i R_i Y_i}{\tilde{p}_j}.$$

Reduces the effective dimension of $(p_1, \ldots, p_B)$; a Hájek-binned version is immediate.

Bayesian answer II: Sieve



Chris Sims (Oct 21, 1942 – Mar 14, 2026)

Bayesian sieve [Sims, 2010, 2012]

Acknowledge the infinite-dimensional unknown: break the range of $p(\cdot)$ into k segments on which θ and p are independent; estimate θ by a step function. The induced law on the p_j converges to a Dirichlet, making the posterior mean of ψ tractable.

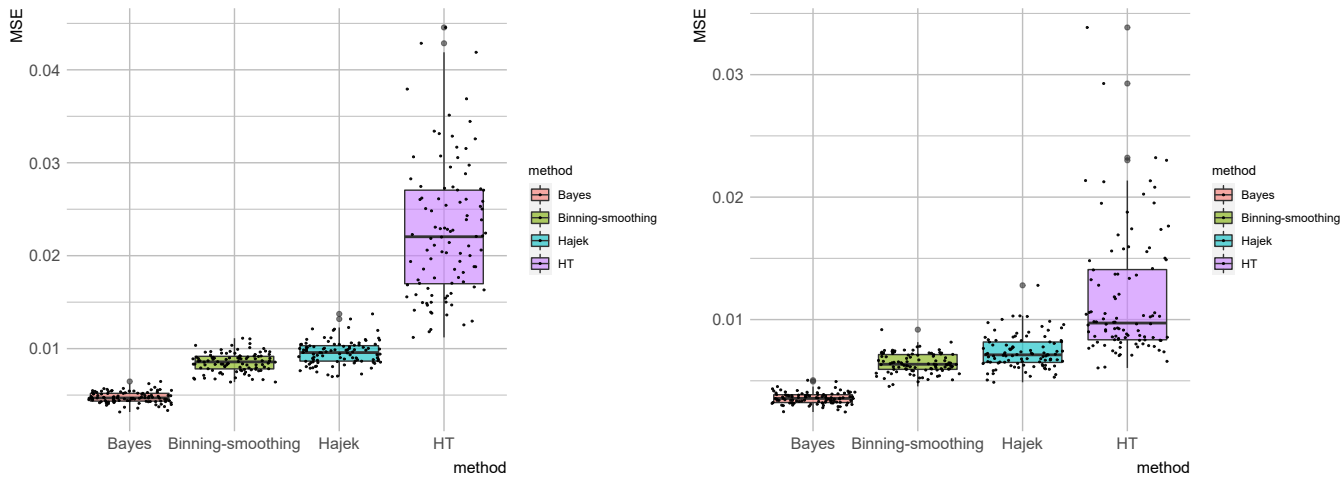
Also: a Gaussian-likelihood variant [Harmeling and Touissant, 2007]; all trade a vanishing bias for lower variance.

Simulation: MCAR: Bayes $<$ Binning $<$ Hájek $<$ HT

$B = 1000$, $\delta = 0.01$, $n = 100$; $\theta_b \sim \mathcal{U}[a, b]$. MSE $\times 10^2$ over 100 replicates.

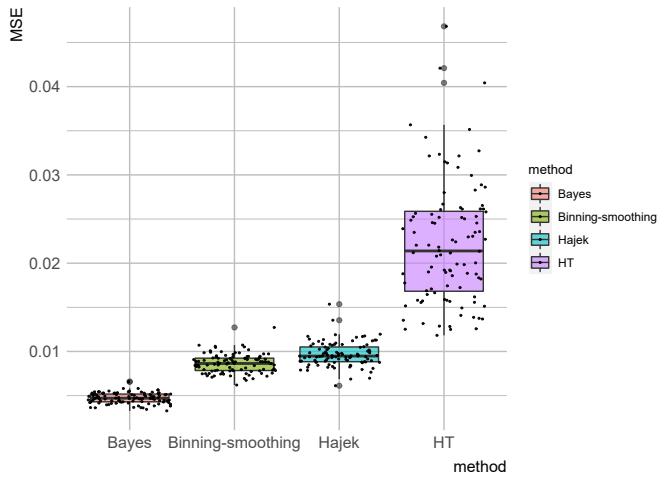
	Support $[a, b]$ of θ			
Method	$[0.6, 0.9]$	$[0.1, 0.9]$	$[0.1, 0.4]$	$[0.35, 0.65]$
Bayes (Li)	**0.372**	**0.475**	**0.357**	**0.473**
Binning–smoothing	0.640	0.852	0.644	0.865
Hájek	0.718	0.958	0.735	0.967
Horvitz–Thompson	3.090	2.271	1.175	2.222

- Bayes (Li) uniformly lowest MSE; HT uniformly worst.
- Binning–smoothing and Hájek occupy the middle ground: achieve variance reduction *without* a substantial bias.

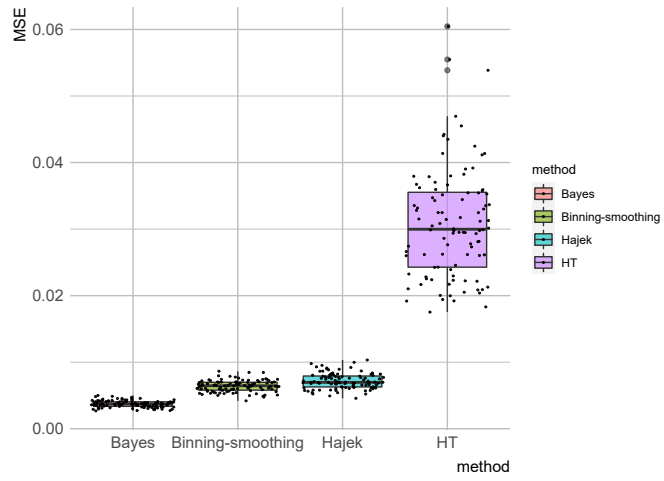


(a) $\theta_b \sim \mathcal{U}(0.1, 0.9)$

(b) $\theta_b \sim \mathcal{U}(0.1, 0.4)$



(c) $\theta_b \sim \mathcal{U}(0.35, 0.65)$

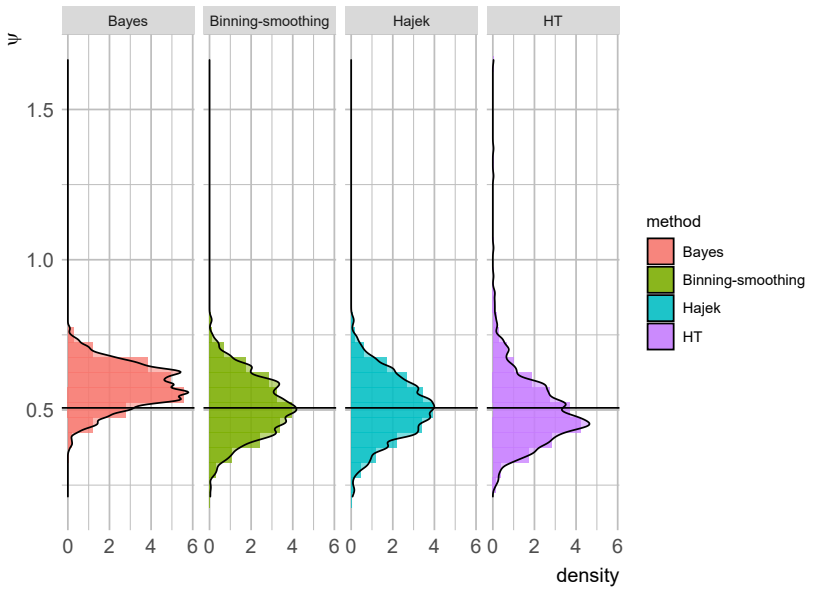


(d) $\theta_b \sim \mathcal{U}(0.6, 0.9)$

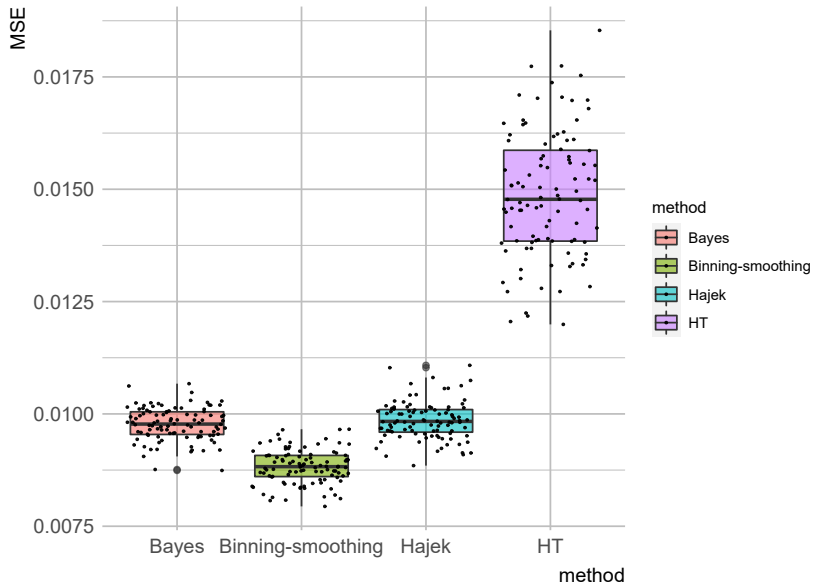
Figure 2: MSE comparison for Horvitz-Thompson, Hájek and the Li's estimator : Bayes' estimator has the lowest MSE, followed by Hájek, and HT has the largest MSE.

Simulation: MAR: the Bayes bias appears

Now let missingness confound the outcome: $\theta_b = \gamma\, p_b + (1-\gamma)\, U_b$, $\gamma = 0.5$, so $\psi = \mathbb{E}(\theta) = 0.5$.



(a) $\theta_b = \gamma \times p_b + (1-\gamma) \times \mathcal{U}(0,1)$



(b) $\theta_b = \gamma \times p_b + (1-\gamma) \times \mathcal{U}(0,1)$

Figure 3: Histogram, Kernel density plots and MSE comparison for Horvitz–Thompson, Hájek and the Bayes estimators for MAR.

Missing at random

What happens

- θ and p positively correlated $\Rightarrow$ **Li's estimator biases upward**.
- Hájek and binned–smoothed are **unaffected**.
- Binning–smoothing has the **lowest MSE** of all four.

Takeaway

The full-Bayes shrinkage estimator is superb under MCAR but *must be used with care* under confounding. Binning–smoothing is the robust all-rounder: it works for both MCAR and MAR.

Application: Average Treatment Effect Estimation

IPW estimators apply naturally to **average treatment effect (ATE)** estimation in the potential-outcomes framework [Delevoye and Sävje, 2020, Khan and Ugander, 2023]; see Rubin [1974], Imbens [2004], Cunningham [2021] for background.

- Following Khan and Ugander [2023]: triplets $(Y_k(0), Y_k(1), p_k)$, $k = 1, \ldots, n$, but we observe only $Y_k(I_k)$ with $I_k \sim \text{Bernoulli}(p_k)$. Target:
$$\tau = \mathbb{E}[Y_k(1) - Y_k(0)].$$
- Estimate the two means $\psi_1 = \mathbb{E}[Y_k(1)]$ and $\psi_0 = \mathbb{E}[Y_k(0)]$ separately, then $\hat{\tau} = \hat{\psi}_1 - \hat{\psi}_0$.
- Each ψ_j is a **survey-sampling** problem: HT, Hájek, or the adaptive IPW estimator of Khan and Ugander [2023] (a Trotter–Tukey construction, Trotter and Tukey 1956) all apply.

Khan and Ugander [2023] minimize the variance of the between-group difference; both the separate and joint AIPW estimators attain lower MSE than HT and Hájek.

Prediction-Powered Inference with IPW

Prediction-powered inference (PPI)

Labeled data $\{(Y_i, X_i)\}_{i=1}^n$ and a predictor $f(X)$; a large unlabeled set $\{\tilde{X}_i\}_{i=1}^N$, $N \gg n$. Estimand $\theta^\star = \mathbb{E}[Y]$.

$$\hat{\theta}_{\text{PPI}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N} f(\tilde{X}_i)}_{\text{predict on large set}} - \underbrace{\frac{1}{n}\sum_{i=1}^{n}\big(f(X_i) - Y_i\big)}_{\text{rectifier } \hat{\Delta}} = \hat{\theta}_f - \hat{\Delta}.$$

- $\mathbb{E}[\hat{\theta}_{\text{PPI}}] = \theta^\star$ regardless of the quality of f; the two terms are independent, so variances add.
- Angelopoulos et al. [2023]: PPI beats classical (labeled-only) inference iff $N \gg n$ *and* $\text{Var}(f(X) - Y) < \text{Var}(Y)$.

Key idea: the PPI rectifier is a Hájek estimator

Write residuals $e_i = f(X_i) - Y_i$ and let R_i indicate a label, $\mathbb{P}(R_i{=}1) = \xi_i$.

Under simple random sampling ($\xi_i \equiv \xi$)

$$\frac{1}{n_{\text{lab}}} \sum_{i:R_i=1} (f(X_i) - Y_i) \;=\; \frac{\sum_i R_i e_i / \xi}{\sum_i R_i / \xi} \;=\; \text{Hájek estimator of } \mathbb{E}[e].$$

- So the standard PPI correction is **exactly** a Hájek estimator applied to prediction residuals. Perhaps, PPI can admit another look from the design-based survey sampling PoV?
- Precursors: the ratio estimator as a Bayes posterior mean [Royall and Pfeffermann, 1982]; "to weight or not to weight" [Smith, 1988]; the GREG estimator [Cassel et al., 1976].

Informative labeling $\Rightarrow$ IPW-adjusted PPI

When $\mathbb{P}(R_i{=}1 \mid X_i) = \xi_i$ is **not** constant (e.g. younger units over-sampled), the unweighted rectifier is biased. Replace it with an IPW rectifier:

$$\delta_{\mathrm{HT}} = \frac{1}{N}\sum_{i=1}^{N} \frac{R_i}{\xi_i} e_i, \qquad \delta_{\mathrm{Há}} = \frac{\sum_i (R_i/\xi_i) e_i}{\sum_i R_i/\xi_i}, \qquad \hat{\theta}_{\mathrm{PPI},\bullet} = \frac{1}{N}\sum_i f(\tilde{X}_i) - \delta_\bullet.$$

- HT form: design-unbiased given correct ξ_i. Hájek form: approximately unbiased, usually lower variance.
- In practice ξ_i is *estimated* (e.g. logistic $R_i \mid X_i$); under a correctly specified model, plugging in $\hat{\xi}_i$ has negligible effect on bias, coverage and efficiency.
- Taxonomy: this is **MAR**; the usual PPI setup is MCAR, where HT and Hájek coincide.

Variance, cross-fitting, and Binning-Smoothing

Cross-fitting [Zrnic and Candès, 2024]: fit $f^{(-k)}$ out of fold to remove the reuse bias.

Binning–smoothing [Ghosh, 2015] carries over: bin the ξ_i, replace by bin midpoints, average the within-bin HT means. Similar to the binning-smoothing idea as before.

Variance / CI (independent Bernoulli labeling):

$$\widehat{\mathrm{Var}}(\delta_{\mathrm{HT}}) = \frac{1}{N^2} \sum_{i=1}^{N} \frac{1-\hat{\xi}_i}{\hat{\xi}_i^2} R_i e_i^2,$$

$$\hat{\theta}_{\mathrm{PPI,HT}} \pm z_{1-\alpha/2} \sqrt{\widehat{\mathrm{Var}}(\delta_{\mathrm{HT}})}.$$

The Hájek form uses a delta-method (centered-residual) variance.

When f is trained independently of the labeled sample, the prediction term is $O(N^{-1})$ and negligible, so $\mathrm{Var}(\hat{\theta}_{\mathrm{PPI,HT}}) \approx \mathrm{Var}(\delta_{\mathrm{HT}})$.

Real data: mean BMI from NHANES (informative labeling by age)

Treat NHANES 2013–14 as the finite population; label with $\xi_i = \sigma(3 - 0.05\,\mathrm{Age}_i)$; predict BMI by linear regression. True mean BMI $= 27.04$.

Estimator	Empirical bias	95% CI width
Classic (labeled-only)	−0.628	0.396
Horvitz–Thompson	−0.111	0.229
Hájek	−0.028	0.229
PPI (unweighted)	−0.024	0.333
PPI (weighted, Hájek)	0.010	0.328

- The classic mean is badly biased; HT partially corrects; Hájek and both PPI variants nearly eliminate the bias.
- Weighted PPI is *nearly unbiased* at a modest cost in interval width: a bias–variance trade-off?

Synthetic study: coverage under informative labeling

$N = 500$, binary outcome, $\xi_i = \text{logit}^{-1}(0.5X_i)$, $\hat{\xi}_i$ from logistic regression; 200 replicates. True mean $= 0.49$.

Estimator	Estimate	Bias	Width	Coverage
Classic	0.543	0.0529	0.123	0.660
Horvitz–Thompson	0.491	0.0012	0.092	0.965
Hájek	0.491	0.0012	0.092	0.965
PPI (unweighted)	0.498	0.0077	0.120	1.000
PPI (weighted)	0.491	0.0012	0.092	0.975

- Classic: 66% coverage: naïve inference fails under informative labeling.
- Weighted PPI matches HT/Hájek efficiency *and* restores nominal coverage. Across $p_{\text{lab}} \in \{0.01, 0.02, 0.05\}$ (with N=10,000) it consistently has the lowest bias.

Evidence Estimation: IPW in Disguise

Importance sampling is inverse probability weighting

To estimate $\psi = \mathbb{E}_F\{I(X)\} = \int I(x)\, dF(x)$ with proposal g:

$$\hat{\psi}_{\mathrm{IS}} = \frac{1}{n}\sum_{i=1}^{n} I(x_i)\frac{f(x_i)}{g(x_i)} = \frac{1}{n}\sum_{i=1}^{n} I(x_i)\, w(x_i).$$

- This is the **HT estimator**: the "inclusion probability" is g, the weight is $w = f/g$.
- The self-normalized (ratio) version $\sum_i I(x_i)w(x_i) / \sum_i w(x_i)$ is the **Hájek estimator** [Hesterberg, 1988, Firth, 2011].
- Poor $g \Rightarrow$ heavy-tailed weights $\Rightarrow$ infinite variance: Basu's circus, again.

Digression: the harmonic mean estimator

Marginal likelihood (evidence) $Z = \int P(x \mid \theta)\, P(\theta)\, d\theta$.

Newton and Raftery [1994] take the special case $g =$ posterior: draw $\theta_1, \ldots, \theta_n$ from the posterior and *invert* the mean reciprocal likelihood,

$$\widehat{Z}_{\mathrm{HM}} = \left[\frac{1}{n} \sum_{i=1}^{n} \frac{1}{P(x \mid \theta_i)} \right]^{-1}.$$

- Importance sampling with proposal $g =$ posterior: the weights $w_i = 1/P(x \mid \theta_i)$ are *unbounded* where the likelihood is small.
- Consistent by the LLN, but the weights are heavy-tailed $\Rightarrow$ typically *infinite variance* [Neal, 1994]. Basu's circus, once more.

Digression (contd.): Worst Monte Carlo Method Ever?



The Harmonic Mean of the Likelihood: Worst Monte Carlo Method Ever

The method is very simple. You obtain a sample of points from the posterior distribution of the model given the observed data, probably using MCMC. For each such point, you compute the reciprocal of the likelihood, average these reciprocal likelihood values, and then take the reciprocal of the average as an estimate of the marginal likelihood of the model. That is, given a sample $t_1, \ldots, t_n$ from the posterior distribution, the marginal likelihood is estimated by

$$P(x) \approx 1 \Big/ \left[(1/n) \sum_{i=1}^{n} 1/P(x|t_i)\right]$$

The good news is that the Law of Large Numbers guarantees that this estimator is consistent — ie, it will very likely be very close to the correct answer if you use a sufficiently large number of points from the posterior distribution. To see this, note that the expectation of what is averaged is

$$\int \frac{1}{P(x|t)} P(t|x)\, dt = \int \frac{1}{P(x|t)} \frac{P(x|t)P(t)}{P(x)}\, dt = \frac{1}{P(x)} \int P(t)\, dt = \frac{1}{P}$$

The bad news is that the number of points required for this estimator to get close to the right answer will often be greater than the number of atoms in the observable universe. The even worse news is that it's easy for people to not realize this, and to naively accept estimates that are nowhere close to the correct value of the marginal likelihood.

Neal [1994]: because $\widehat{Z}_{\mathrm{HM}}$ sees the prior only through the posterior, it barely moves even when the true evidence shifts by orders of magnitude.

From naive averaging to ordered quadrature

Naive MC averages the *unordered* draws,

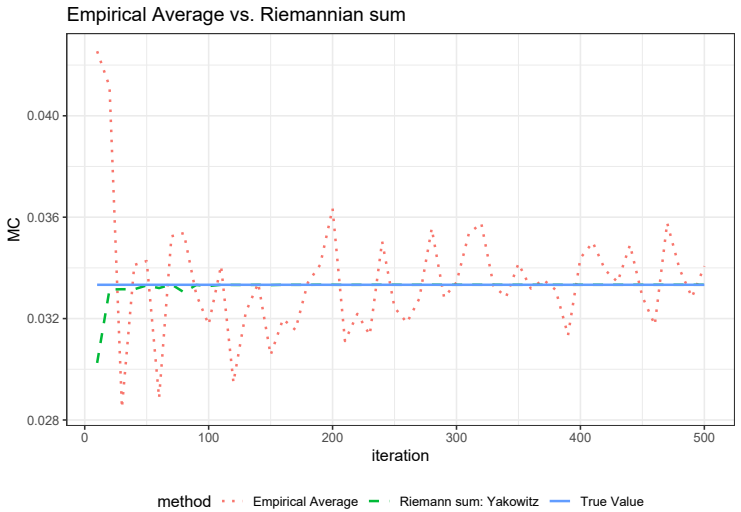
$$\hat{\psi}_{\mathrm{IS}} = \frac{1}{n}\sum_i \frac{L(x_i)f(x_i)}{g(x_i)}, \qquad O(n^{-1}).$$

Reorder the draws, apply a **quadrature** rule:

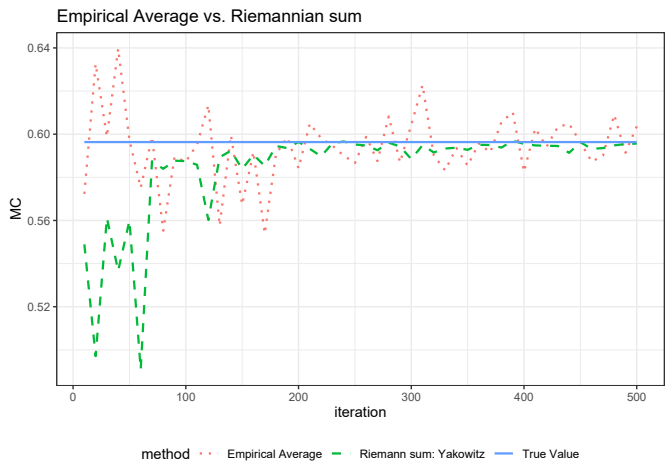
- **Yakowitz (1978)**: Using a trapezoid rule on ordered *uniforms* for $\psi = \int_0^1 L(x)dx$: rate $O(n^{-4})$.
- Extended to general f via $x_{(i)} = F^-(u_{(i)})$,
- Riemann sum:

$$\sum_i L(x_{(i)})f(x_{(i)})(x_{(i+1)} - x_{(i)}),$$

rate $O(n^{-2})$ and bias $O(n^{-1})$. [Philippe, 1997, Philippe and Robert, 2001]



Estimating $\mathrm{Beta}(3,3)$: ordered Riemann sum vs. naive average.



Estimating $\mathbb{E}(1/(1+U))$ for $U \sim \mathrm{Exp}(1)$

Riemann estimators

- Despite the much faster convergence rates, the Yakowitz estimator has limited use.
- Extending trapezoid rules to higher dimension would be thwarted by the 'curse of dimensionality'.
- An overlooked fact was that high-dimensional integrals can be written as uni-dimensional integrals over the positive real line or over the unit interval (Nested Sampling).
- **Nested sampling** is a perfect situation to invoke the Yakowitz estimator.

Riemann sum estimators [Yakowitz et al., 1978]

$+$

Nested sampling [Skilling, 2006]

$\Downarrow$

Quantile Importance Sampling: QIS

Quantile importance sampling

Every evidence integral is *one-dimensional*. With $Z(y) = \mathbb{P}\{L(\mathbf{X}) > y\}$ and $\Lambda = Z^{-1}$, the **Lorenz identity** gives

$$\psi = \int_0^\infty Z(y)\, dy = \int_0^1 \Lambda(s)\, ds, \qquad \Lambda(U) \overset{\mathrm{D}}{=} L(\mathbf{X}), \quad Z(L(\mathbf{X})) \sim \mathcal{U}(0,1).$$

Apply the Yakowitz trapezoid to Λ at ordered uniforms ($U_{(0)}$=0, $U_{(n+1)}$=1):

$$\hat{\psi}_{\mathrm{QIS}} \;=\; \tfrac{1}{2} \sum_{i=1}^{n+1} (U_{(i)} - U_{(i-1)}) \{\Lambda(U_{(i-1)}) + \Lambda(U_{(i)})\}$$

The trick

Estimate $\Lambda(u_{(i)})$ by the sample quantile $Y_{\lceil m \times u_{(i)} \rceil}$ of $\{L(\mathbf{x}_j)\}_{j=1}^m$, $\mathbf{x}_j \sim$ prior: same as reordering the likelihood-ordinate quantiles onto $[0,1]$.

QIS: the $O(n^{-4})$ rate

Main result

If Λ'' is bounded on $[0,1]$, then for some M and all $n \geq 1$,

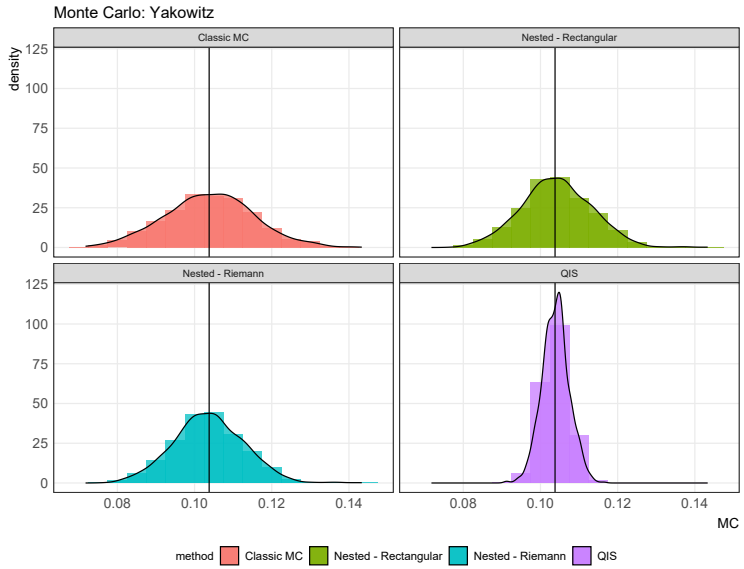
$$\mathbb{E}[(\psi - \hat{\psi}_{\text{QIS}})^2] \leq M/n^4.$$

The fast trapezoid rate now holds for evidence integrals in *any* dimension.

Multivariate t $(d = 50, \nu = 2)$, true $\psi = U(26, 2, 1) = 1.95 \times 10^{-29}$:

$$\underbrace{2.96 \times 10^{-29}}_{\text{QIS}} \quad \text{vs.} \quad \underbrace{2.0 \times 10^{-31}}_{\text{naive}}$$

right order of magnitude vs. off by $\sim 100\times$.



QIS (ordered uniform) vs. Skilling's $e^{-i/N}$ vs. naive MC.

Envoi

Connections

1. Basu showed design-based IPW can be *unbiased yet absurd*.
2. The weak paradoxes (RRW) sharpen this into a high-dimensional challenge to Bayes.
3. Exchangeability / **Stein shrinkage** [Li, 2010], binning–smoothing [Ghosh, 2015], and the sieve [Sims, 2012] propose resolutions: lower variance at a controlled bias.
4. The *same* Hájek normalization can be applied to modern **prediction-powered inference** under informative labeling.
5. Also in evidence estimation: importance sampling can be thought of as IPW, with quadrature-style corrections.

Takeaway: Inverse probability weighting is still important, across survey sampling, interpretable ML, and Bayesian computation. It is also important to think about *weak paradoxes* in our foundation.

Thank you!

The talk was based on these three inter-connected ideas.

- **IPW: from survey sampling to evidence estimation (New England Journal of Statistics and Data Science. 2026).** arXiv:2204.14121
- **Prediction-powered inference with IPW (Statistical Science Short Communications 2026).** https://arxiv.org/abs/2508.10149
- **Quantile importance sampling (Brazilian Journal of Probability & Statistics. 2025).** arXiv:2305.03158

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Backup: Li's estimator: variance approximation

With $\bar{\theta}_{\cdot p} = \frac{1}{B}\sum_b \theta_b p_b$, the delta method gives

$$\mathbb{V}(\hat{\psi}_{\mathrm{Li}}) \approx \left(\tfrac{n\bar{\theta}_{\cdot p}+1}{n\bar{p}+2}\right)^2 \left[\frac{n(\bar{\theta}_{\cdot p} - \bar{\theta}^2_{\cdot p})}{(n\bar{\theta}_{\cdot p}+1)^2} + \frac{n(\bar{p}-\bar{p}^2)}{(n\bar{p}+2)^2} - \frac{2n\bar{\theta}_{\cdot p}(1-\bar{p})}{(n\bar{p}+2)(n\bar{\theta}_{\cdot p}+1)}\right].$$

Unlike $\mathbb{V}(\hat{\psi}_{\mathrm{HT}}) = \frac{1}{n}\{\frac{1}{B}\sum_b \theta_b/p_b - \psi^2\}$, there are *no* individual p_b in a denominator, so it does not inflate as $p_b \to 0$.

Backup: sparsity via global-local shrinkage

For sparse $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_B)$, augment Li's model with a horseshoe-type prior:

$$\theta_b \mid \alpha_T, \psi \overset{ind}{\sim} f_{\mathrm{HS}}(\theta; \alpha_T \psi, \alpha_T(1-\psi), \tau), \qquad (\psi, \alpha_T \mid \alpha_F) \sim \mathrm{Beta}(\alpha_F, \alpha_F) \times \pi(\alpha_T),$$

with τ a global shrinkage parameter adapting to sparsity: tail-robust regularization for treatment-effect estimation with more controls than observations.

Quantile Importance Sampling

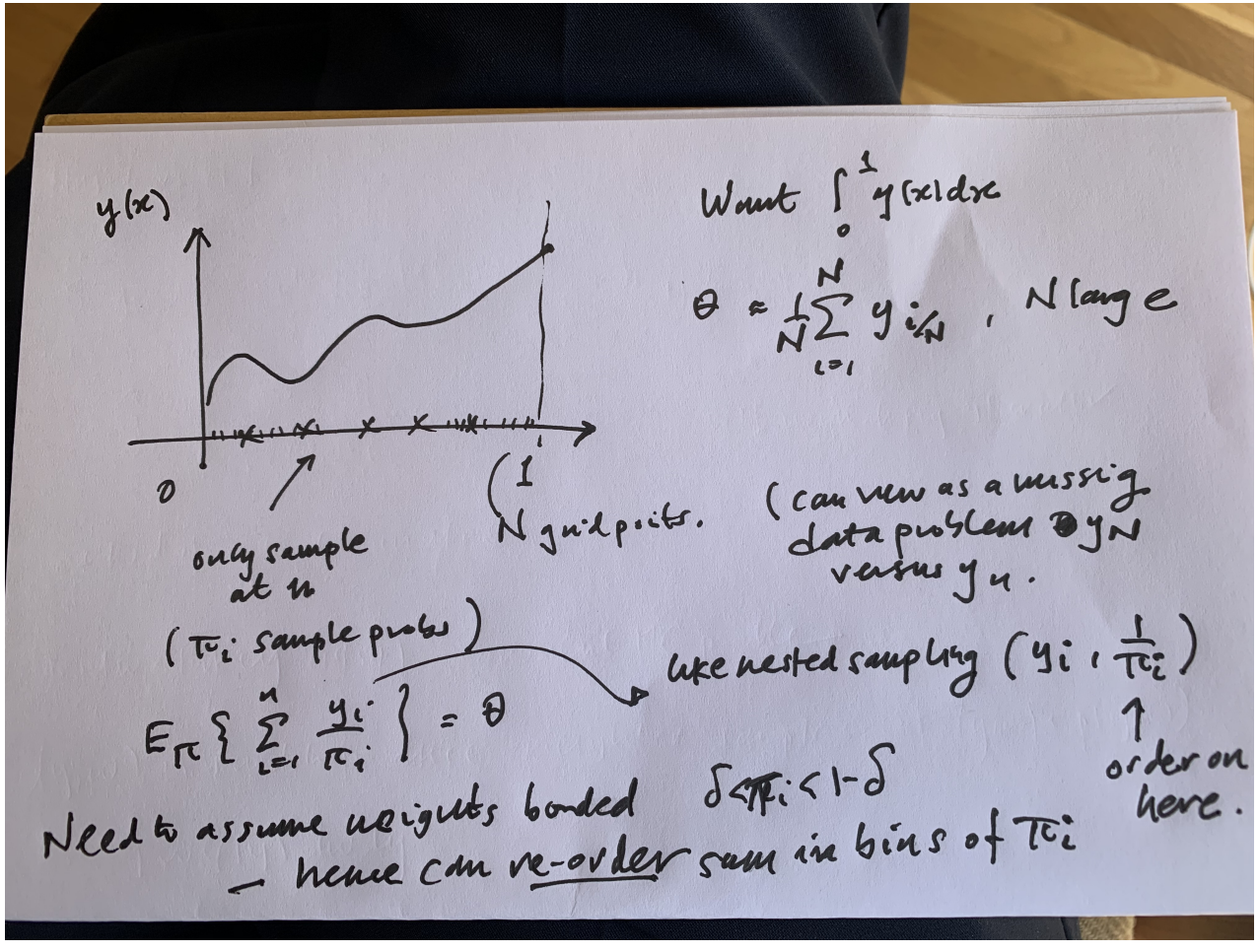


Figure 4: Idea!

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